ARBITRARY POINTED MAPS BETWEEN POSTNIKOV TRIPLES

Purpose. The informal proof that is formalised, step by step (each statement names its declarations; Section 9 gives the correspondence), of the classification of *arbitrary* pointed maps between pointed simply connected 3-types by maps of Postnikov triples (Theorem 3), the induced equivalence of homotopy categories (Section 3), and the reformulation of the two sphere arguments through it (Sections 5, 6). The existing development (`sag/pi3-s2-postnikov`) classifies the *objects* (`PostnikovClassification.classification`) and proves specialised mapping theorems (`MapFib`, `MarkDirect`); it does not classify arbitrary maps, and those endpoints are not used as the reason for the results below.

Attribution. The strategy of identifying the sphere universal objects on the triple side, and the embedding of pointed 2-connected 4-types into pointed simply connected 3-types by looping, are David Wärn's, as communicated by the user; the delooping machinery (`DeloopMaps`, Corollary 11) is his. The object classification is Hart–Milner's (Owen Milner's equivalence). We do not know Owen Milner's (unpublished) treatment of maps and do not claim to reproduce it.

Conventions. As in the development, $\mathsf{isConn}\,n$ means $(n-2)$-connected and $\mathsf{isTrunc}\,n$ means $(n-2)$-truncated. Composition is written in diagrammatic order: $f\,;g$ is "f, then g" (`ptcomp`). For pointed $f=(f_1,f_0)$, g: $(f\,;g)_1 = g_1\circ f_1$ and $(f\,;g)_0 = \mathsf{ap}_{g_1}(f_0)\cdot g_0$. Path composition $p\cdot q$ is `trans`, by J on q, so $p\cdot\mathsf{refl}\equiv p$. $K(A,n)$ is `EMMaps.KK[A].Kn`; $K(h,n)$ is the locked `H4Bilinear.Lev[A,B].Khn`. All types live in U_l unless said otherwise; groups are elements of `Ab` $: U_{l+1}$.

1. The statement

1.1. **Triples and their realisation.** A triple is $t=(A,B,k)$ with $k : K(A,2)\to_* K(B,4)$ (`PostnikovClassification.Triples`). Fix triples $t=(A,B,k)$, $t'=(A',B',k')$ and write

$$V=K(A,2),\ Y=K(B,4),\ V'=K(A',2),\ Y'=K(B',4),$$
$$P_x=(k_1x=y_0),\qquad \mathrm{Fib}k=\textstyle\sum_{x:V}P_x,$$

pointed by $w_0=(v_0,k_0)$, with projection $p=\mathsf{fpr}:\mathrm{Fib}k\to_* V$ (`PostnikovRealization.Triple.F`, `f0`). The facts used: V is $\mathsf{isConn}\,3$ (Lemma 12 uses only that it is 0-connected) and $\mathsf{isTrunc}\,4$ (in fact a 2-type); Y is $\mathsf{isConn}\,5$; Y' is $\mathsf{isTrunc}\,6$; every P_x is $\mathsf{isConn}\,4$ (`MapFib.FibP.cp`, `Connected.conn-id`) and $(y=y')$ is $\mathsf{isTrunc}\,5$ in Y'.

1.2. **Morphisms.**

Definition 1 (`TripleMapsEquiv.Maps2.Mor`)**.**

$$\mathsf{Mor}(t,t')=\sum_{a:\mathrm{Hom}(A,A')}\ \sum_{b:\mathrm{Hom}(B,B')}\big(K(a,2)\,;k'=_{\mathrm{Map}_*(V,Y')}k\,;K(b,4)\big).$$

Accessors `mor-a`, `mor-b`, `mor-H`. The third field is a *path of pointed maps*: a homotopy together with its coherence at the base point.

Definition 2 (`TripleMapsEquiv.Maps2.realise-map`)**.** $\mathsf{realise\text{-}map}(a,b,H)$ = `FibMap.fm` for the square $f=k$, $f'=k'$, $u=K(a,2)$, $v=K(b,4)$, $sq=H$. Explicitly, with $hs_x=\mathtt{happly}(\mathsf{ap}_{(-)_1}H)_x : k'_1(u_1x)=v_1(k_1x)$ and $c=\mathtt{pt\text{-}coh}(H):\mathsf{ap}_{k'_1}(u_0)\cdot k'_0=hs_{v_0}\cdot(\mathsf{ap}_{v_1}k_0\cdot v_0)$,

$$\mathsf{realise\text{-}map}(a,b,H)(x,q)=(u_1x,\ hs_x\cdot(\mathsf{ap}_{v_1}q\cdot v_0)),$$

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pointed by `Sigma-path`$(u_0,$ `fm-pt-q`$)$.

Theorem 3 (`TripleMapsEquiv.Maps2.realise-map-equiv`)**.** *For all triples t, t',* realise-map : $\mathsf{Mor}(t,t') \to \mathrm{Map}_*(\mathrm{Fib}k, \mathrm{Fib}k')$ *is an equivalence. No sphere axiom, no hypothesis on A, B, A', B', k, k'.*

1.3. **Sanity checks.** *Levels.* $\mathrm{Map}_*(V,V') \simeq \mathrm{Hom}(A,A')$ and $\mathrm{Map}_*(Y,Y') \simeq \mathrm{Hom}(B,B')$ are sets (Lemma 14); $\mathrm{Map}_*(V,Y')$ is $\mathsf{isTrunc}\,4$ (a 2-type) by `Lemma4.cor5-map` (V is $\mathsf{isConn}\,3$, Y' is $\mathsf{isTrunc}\,6$), so its identity types are 1-types and $\mathsf{Mor}(t,t')$ is a 1-type. On the other side $\mathrm{Map}_*(\mathrm{Fib}k, \mathrm{Fib}k')$ is a 1-type by `cor5-map` (Fib$k$ is $\mathsf{isConn}\,3$, Fibk' is $\mathsf{isTrunc}\,5$). Neither is a set in general.

The homotopy is data. For $t = (A,0,0)$, $t' = (0,C,0)$: $\mathrm{Hom}(A,0)$ and $\mathrm{Hom}(0,C)$ are contractible and $\mathsf{Mor}(t,t') \simeq (0 =_{\mathrm{Map}_*(K(A,2),K(C,4))} 0) \simeq \mathrm{Map}_*(K(A,2),K(C,3))$, while $\mathrm{Fib}(0) \simeq K(A,2)$ and $\mathrm{Fib}(0 : K(0,2) \to K(C,4)) \simeq \Omega K(C,4) \simeq K(C,3)$, so both sides are $\mathrm{Map}_*(K(A,2),K(C,3))$, whose components are $H^3(K(A,2);C)$, not zero in general. The coefficients alone would give a point. (Not formalised as a separate statement.)

Why the formula is right. Classically it is the statement that maps of two-stage Postnikov systems in this range are maps of bases, of fibres and a homotopy, which follows from the Serre (or dual Blakers–Massey) estimate on the cofibre of $\mathrm{Fib}k \to K(A,2)$; we do not use cofibres (no higher inductive types), but the argument below is a direct, fibrewise replacement for that estimate.

2. Proof of Theorem 3

The proof is a chain of equivalences whose composite is, by definition, realise-map. Write, for a pointed $c = (c_1, c_0) : V \to_* Y'$ and $Q^c_x = (c_1 x = y'_0)$:

$$T(c) = \sum_{F : \prod_x P_x \to Q^c_x} \big(F_{v_0} k_0 = c_0\big), \qquad S(c) = \sum_{\beta : \mathrm{Map}_*(Y,Y')} \big(c =_{\mathrm{Map}_*(V,Y')} k \,;\, \beta\big).$$

$T(c)$ is the type of pointed null-homotopies of $p;c$ in fibrewise form; $S(c)$ is the type of factorisations of c through k.

2.1. **Maps out of** Fib$k$ **into** Fibk'**.**

Lemma 4 (`TripleMapsEquiv.Dec.dec-equiv`)**.** *The map*

$$\mathrm{Dec} : \sum_{\alpha : \mathrm{Map}_*(V,V')} T(\alpha \,;\, k') \to \mathrm{Map}_*(\mathrm{Fib}k, \mathrm{Fib}k'),$$
$$(\alpha, F, e) \mapsto \big((x,q) \mapsto (\alpha_1 x, F_x q),\ \textit{Sigma-path}(\alpha_0, \pi(e))\big)$$

is an equivalence, where for $e : F_{v_0} k_0 = \mathsf{ap}_{k'_1} \alpha_0 \cdot k'_0$ (this is $(\alpha \,;\, k')_0$ by definition), $\pi(e) : \alpha_{0*}(F_{v_0} k_0) = k'_0$ is ***transport-fib****, then $\mathsf{ap}_{(\mathsf{ap}_{k'_1} \alpha_0)^{-1} \cdot -}(e)$, then* ***trans-cancel*** *(exactly the steps of* ***FibMap. fm-pt-q****).*

Proof. (i) Precomposition with p, $(V \to V') \to (\mathrm{Fib}k \to V')$, is an equivalence: `ConnectedMaps.Elim` at $m = \mathsf{isConn}\,4$, along p whose fibres are $\mathsf{isConn}\,4$ (`FibP.cp`), into the constant family V', which is $\mathsf{isTrunc}\,4$.

(ii) The map $(a_1, F) \mapsto ((x,q) \mapsto (a_1 x, F_x q))$,

$$\textstyle\sum_{a_1 : V \to V'} \prod_x P_x \to (k'_1(a_1 x) = y'_0) \ \to\ (\mathrm{Fib}k \to \mathrm{Fib}k'),$$

is an equivalence: it is the composite of currying $\prod_x \prod_{q : P_x} \simeq \prod_{w : \mathrm{Fib}k}$ (inverse by definition), `sigma-base` along (i), and the distributivity $\sum_{g : \mathrm{Fib}k \to V'} \prod_w R(g\,w) \simeq (\mathrm{Fib}k \to \sum_y R\,y)$ (inverse by definition), for $R\,y = (k'_1 y = y'_0)$.

(iii) $\mathrm{Map}_*(\mathrm{Fib}k, \mathrm{Fib}k') = \sum_G (G\,w_0 = w_0')$; pulling back along (ii) (`sigma-base`), the fibre over (a_1, F) is $(a_1 v_0, F_{v_0} k_0) = w_0'$ *by definition* of the map (ii).

(iv) For fixed (a_1, F), the map $(\alpha_0, e) \mapsto \mathtt{Sigma\text{-}path}(\alpha_0, \pi(e))$, $\sum_{\alpha_0 : a_1 v_0 = v_0'} (F_{v_0} k_0 = \mathsf{ap}_{k_1'} \alpha_0 \cdot k_0') \to \big((a_1 v_0, F_{v_0} k_0) = w_0'\big)$, is an equivalence: π is a composite of fixed paths (a left multiplication, `Fibres.trans-equiv`, and a right multiplication, `rmul-equiv`) with `ap` of the equivalence $(\mathsf{ap}_{k_1'} \alpha_0)^{-1} \cdot -$ (`ClassBase.ap-equiv`), so a fibrewise equivalence (`Fibres.sigma-equiv`), followed by `Sigma-path`, the inverse of `ConnectedMaps.sigma-paths` (`Whitehead.inv-equiv`).

Reordering the Σ-components ($\sum_\alpha \sum_F \sum_e \simeq \sum_{a_1} \sum_F \sum_{\alpha_0} \sum_e$, inverse by definition) and composing (ii)–(iv) gives Dec. □

2.2. **The key lemma: null-homotopies are factorisations.** For $\beta : \mathrm{Map}_*(Y, Y')$ write $G_y^\beta(q) = \mathsf{ap}_{\beta_1} q \cdot \beta_0 : \beta_1 y = y_0'$ for $q : y = y_0$ (so $(k\,;\beta)_0 = G^\beta_{k_1 v_0}(k_0)$), and $Z_x^\beta = (c_1 x = \beta_1(k_1 x))$.

Definition 5 (Φ, `TripleMaps.Key.Phi`)**.** $\Phi_c : S(c) \to T(c)$, $\Phi_c(\beta, s) = \big(x\,q \mapsto hs_x \cdot G^\beta_{k_1 x}(q),\ (\mathtt{pt\text{-}coh}\, s)^{-1}\big)$, with $hs = \mathtt{happly}(\mathsf{ap}_{(-)_1} s)$ and $\mathtt{pt\text{-}coh}\, s : c_0 = hs_{v_0} \cdot (k\,;\beta)_0$.

Lemma 6 (`TripleMaps.Key.key`)**.** *For every pointed $c : V \to_* Y'$, Φ_c is an equivalence.*

The proof occupies the rest of this subsection. For $x : V$, $F_x : P_x \to Q_x^c$ and β, put

$$\mathsf{Tr}_x^\beta(F_x) = \sum_{z : Z_x^\beta} \big((q \mapsto z \cdot G^\beta_{k_1 x} q) =_{P_x \to Q_x^c} F_x\big) \qquad (\text{“}F_x \text{ is a translate of } G^\beta\text{”}).$$

Lemma 7 (paths of pointed maps; `TripleMaps.PtPath.equiv`)**.** *For pointed $c, g : X \to_* Y$, $s \mapsto (\mathtt{happly}(\mathsf{ap}_{(-)_1} s), \mathtt{pt\text{-}coh}\, s)$,*

$$(c = g) \to \textstyle\sum_{h : \prod_x c_1 x = g_1 x} (c_0 = h_{x_0} \cdot g_0),$$

is an equivalence.

Proof. Fundamental theorem of identity types (`FundThm.FT`): the family on the right is pointed at $g = c$ by $(\mathsf{refl},\ (\mathtt{trans\text{-}refl\text{-}l})^{-1})$, which is the value of the map at refl (both by J), and its total space $\sum_g \sum_h (c_0 = h_{x_0} \cdot g_0)$ is contractible: $\sum_{g_1} \sum_h$ is the singleton of c_1 (function extensionality), and then $\sum_{g_0} (c_0 = \mathsf{refl} \cdot g_0) \simeq \sum_{g_0} (c_0 = g_0)$ is a singleton. The map so obtained by J agrees with $(\mathtt{happly} \circ \mathsf{ap}, \mathtt{pt\text{-}coh})$ by J on s. □

Lemma 8 (translates; `TripleMaps.Transl.equiv`, `Transl.prop`)**.** *For each x, $z \mapsto (q \mapsto z \cdot G^\beta_{k_1 x} q,\ (z, \mathsf{refl}))$ is an equivalence $Z_x^\beta \simeq \sum_{F_x} \mathsf{Tr}_x^\beta(F_x)$, and $\mathsf{Tr}_x^\beta(F_x)$ is a proposition.*

Proof. The inverse is $(F_x, z, E) \mapsto z$; one round trip is refl, the other is J on E. For the proposition: $\mathsf{Tr}_x^\beta(F_x)$ is the fibre over F_x of $\tau : z \mapsto (q \mapsto z \cdot G^\beta q)$, and τ is the constant embedding $Z_x^\beta \to (P_x \to Z_x^\beta)$ (`FibreH3.const-emb`: P_x is $\mathsf{isConn}\,4$, Z_x^β is $\mathsf{isTrunc}\,5$) followed by postcomposition with the fibrewise equivalence $d \mapsto (q \mapsto d\,q \cdot G^\beta q)$ (`ConnectedMaps.pi-equiv` of right multiplication by a path, `TripleMaps.rmul-equiv`, by J on the path); `ap` of the composite is the composite of the `ap`'s (`LoopCalc.ap-ap`) and `ap` of an equivalence is an equivalence (`ClassBase.ap-equiv`); an embedding has propositional fibres (`Groupoid.emb-fib-prop`). □

Lemma 9 (`TripleMaps.Key.fac-equiv`)**.** *For fixed β, the map*

$$(c = k\,;\beta) \to \sum_{(F,e) : T(c)} \prod_x \mathsf{Tr}_x^\beta(F_x), \qquad s \mapsto (\Phi_c(\beta, s),\ x \mapsto (hs_x, \mathsf{refl}))$$

is an equivalence.

Proof. Lemma 7 at $g = k\,;\beta$ gives $(c = k\,;\beta) \simeq \sum_{h:\prod_x Z_x^\beta}(c_0 = h_{v_0} \cdot G^\beta_{k_1 v_0} k_0)$. Lemma 8 under $\prod_x$ (`pi-equiv`) and distributivity give $\prod_x Z_x^\beta \simeq \sum_F \prod_x \mathsf{Tr}_x^\beta(F_x)$, with $h \mapsto (x\,q \mapsto h_x \cdot G^\beta q,\ x \mapsto (h_x, \mathsf{refl}))$; in particular $F_{v_0}k_0 \equiv h_{v_0} \cdot G^\beta_{k_1 v_0} k_0$ by definition, so the pointing condition is carried by `sigma-base` to $(c_0 = F_{v_0}k_0)$, and `sym` turns it into $F_{v_0}k_0 = c_0$. Reordering the components gives the stated map. □

Definition 10 (restriction to the fibre; `TripleMaps.Key.res`)**.** For $(F, e) : T(c)$, $\mathrm{res}(F, e) : \Omega Y \to_* \Omega Y'$ is $q \mapsto c_0^{-1} \cdot F_{v_0}(k_0 \cdot q)$, pointed by $c_0^{-1} \cdot F_{v_0}k_0 = c_0^{-1} \cdot c_0 = \mathsf{refl}$ (via e; note $k_0 \cdot \mathsf{refl} \equiv k_0$). (This is `FibreH3.FibAtC.res`, conjugated by c_0.)

Lemma 11 (`TripleMaps.Key.to-loop`, `from-loop`)**.** *For $(F, e) : T(c)$ and $\beta : \mathrm{Map}_*(Y, Y')$, the propositions $\mathsf{Tr}^\beta_{v_0}(F_{v_0})$ and $(\mathrm{res}(F, e) =_{\mathrm{Map}_*(\Omega Y, \Omega Y')} \mathsf{loop}\,\beta)$ are logically equivalent.*

Proof. The second is a proposition because $\mathrm{Map}_*(\Omega Y, \Omega Y')$ is a set (`cor5-map`: ΩY is $\mathsf{isConn}\,4$, $\Omega Y'$ is $\mathsf{isTrunc}\,5$). We use the propositional equation $\mathsf{loop}\,\beta(q) = \beta_0^{-1} \cdot (\mathsf{ap}_{\beta_1} q \cdot \beta_0)$ (`Maps.loop-ap`; it is not the definition of loop).

($\Rightarrow$) Let (z, E), so $F_{v_0}w = z \cdot (\mathsf{ap}_{\beta_1} w \cdot \beta_0)$ for all $w : P_{v_0}$ (`happly` E). Since $\Omega(\Omega Y')$ is $\mathsf{isConn}\,2$, pointed maps $\Omega Y \to_* \Omega Y'$ with pointwise equal underlying maps are merely equal (`FibreH3.pi-mere`), and the goal is a proposition, so it suffices that $c_0^{-1} \cdot F_{v_0}(k_0 \cdot q) = \beta_0^{-1} \cdot (\mathsf{ap}_{\beta_1} q \cdot \beta_0)$ for all q. From e and E at $w = k_0$: $c_0 = z \cdot (\mathsf{ap}_{\beta_1} k_0 \cdot \beta_0)$. The claim is then the groupoid identity

$$(z \cdot (\mathsf{ap}_{\beta_1} k_0 \cdot \beta_0))^{-1} \cdot (z \cdot (\mathsf{ap}_{\beta_1}(k_0 \cdot q) \cdot \beta_0)) = \beta_0^{-1} \cdot (\mathsf{ap}_{\beta_1} q \cdot \beta_0),$$

valid for all $k_0 : y = y_0$, z and q (`TripleMaps.grp1`: $(z \cdot A)^{-1} \cdot (z \cdot B) = A^{-1} \cdot B$, by `idJR` on z; `Groupoid.ap-trans`; `TripleMaps.grp3`: $(A \cdot b_0)^{-1} \cdot ((A \cdot M) \cdot b_0) = b_0^{-1} \cdot (M \cdot b_0)$, by J on b_0).

($\Leftarrow$) Let $r : \mathrm{res}(F, e) = \mathsf{loop}\,\beta$. Take $z = F_{v_0}k_0 \cdot (\mathsf{ap}_{\beta_1} k_0 \cdot \beta_0)^{-1}$. For $w : P_{v_0}$ put $q = k_0^{-1} \cdot w$, so $w = k_0 \cdot q$ (`trans-cancel'`); then $F_{v_0}w = c_0 \cdot \mathrm{res}(F, e)(q) = c_0 \cdot \beta_0^{-1} \cdot \mathsf{ap}_{\beta_1} q \cdot \beta_0$ (`happly` r), and with $e : F_{v_0}k_0 = c_0$ the required $z \cdot (\mathsf{ap}_{\beta_1} w \cdot \beta_0) = F_{v_0}w$ follows from `ap-trans`, `tassoc`, `grp3` and `trans-cancel'` (`TripleMaps.Key.goal-at`); it is proved at $w = k_0 \cdot q$ and transported along $k_0 \cdot (k_0^{-1} \cdot w) = w$. Function extensionality gives E. □

Lemma 12 (`TripleMaps.Key.unique-fac`)**.** *For every $(F, e) : T(c)$, $\sum_\beta \prod_x \mathsf{Tr}_x^\beta(F_x)$ is contractible.*

Proof. For each β, $\prod_x \mathsf{Tr}_x^\beta(F_x)$ and $\mathsf{Tr}^\beta_{v_0}(F_{v_0})$ are logically equivalent propositions: one direction is evaluation, the other is `conn-elim` over the 0-connected V into the propositions $\mathsf{Tr}_x^\beta(F_x)$ (Lemma 8; this is `FibreH3.spread`, there stated at $V = K(\mathbb{Z}, 2)$). With Lemma 11, $\sum_\beta \prod_x \mathsf{Tr}_x^\beta(F_x) \simeq \sum_\beta (\mathrm{res}(F, e) = \mathsf{loop}\,\beta)$ (`Logic.sigma-equiv` at the identity), and the latter is Unloop of $\mathrm{res}(F, e)$, contractible by `DeloopMaps.Maps.Over.cor11` at $c = 2$ (Y is $\mathsf{isConn}\,4$, Y' is $\mathsf{isTrunc}\,6$), as in `FibreH3.FibAtC.deloop-c`. □

Proof of Lemma 6. Φ_c is the composite

$$S(c) \xrightarrow{\sum_\beta \text{Lemma 9}} \sum_\beta \sum_{(F,e)} \prod_x \mathsf{Tr}_x^\beta(F_x) \xrightarrow{\text{swap}} \sum_{(F,e)} \sum_\beta \prod_x \mathsf{Tr}_x^\beta(F_x) \xrightarrow{\mathrm{pr}_1} T(c),$$

the swap has inverse by definition, and pr_1 is an equivalence by Lemma 12 (`SphereTruncation.fst-equiv`). The composite is Φ_c by definition. □

Remark 13. This is where the translation data live: a fibrewise map $F_x : P_x \to Q_x^c$ between two "$K(B, 3)$-torsors" is a linear part (the delooping β of its restriction to the fibre over v_0, whose coefficient is b) and a translation (the constant $z = h_x$, the homotopy). The base point fixes the translation at v_0 through e; no point of any other fibre and no representative of a class is chosen.

Lifts are not assumed unique: the ambiguity is exactly the homotopy component h, which is kept as data.

2.3. **Coefficients.**

Lemma 14 (`TripleMapsEquiv.Coef.k2-equiv`, `k4-equiv`)**.** $a \mapsto K(a,2) : \mathrm{Hom}(A,A') \to \mathrm{Map}_*(V,V')$ *and* $b \mapsto K(b,4) : \mathrm{Hom}(B,B') \to \mathrm{Map}_*(Y,Y')$ *are equivalences.*

Proof. Level 2: inverse `H4Bilinear.H2.inv2`, with `k2-surj` and, for $\mathsf{inv2}(K(a,2)) = a$, `k2-inj` applied to `k2-surj`. Level 4: inverse `FibreH3.Climb.inv4` with `surj4`; for $\mathsf{inv4}(K(b,4)) = b$: $\mathsf{loop}\,K(b,m+1) = \mathsf{conj}_m K(b,m)$ (`Lev.loopK`) and $\mathsf{unconj}_m(\mathsf{conj}_m F) = F$ (`TripleMapsEquiv.Coef.unconj-conj`, the computation of `H2.unconj-conj` at level m: $e_A^{-1}\,;e_A\,;F\,;e_B^{-1}\,;e_B = F$), so $\mathsf{inv4}(K(b,4)) = \mathsf{inv3}(K(b,3)) = \mathsf{inv2}(K(b,2)) = b$. □

2.4. **Assembly.** realise-map is the composite

$$\mathsf{Mor}(t,t') \xrightarrow{(a,b,H)\mapsto(a,K(b,4),H)} \textstyle\sum_a S(K(a,2)\,;k') \xrightarrow{\sum_a \Phi} \sum_a T(K(a,2)\,;k')$$
$$\xrightarrow{(a,\theta)\mapsto(K(a,2),\theta)} \textstyle\sum_\alpha T(\alpha\,;k') \xrightarrow{\mathrm{Dec}} \mathrm{Map}_*(\mathrm{Fib}k,\mathrm{Fib}k'),$$

the first and third by `sigma-base` along Lemma 14, the second by Lemma 6 and `Fibres.sigma-equiv`. That the composite *is* `FibMap.fm` holds by definition: the function is $(x,q) \mapsto (u_1 x, hs_x \cdot (\mathsf{ap}_{v_1} q \cdot v_0))$ and the pointing is `Sigma-path`$(u_0, \pi((\texttt{pt-coh}H)^{-1}))$, which is `fm-pt` ($c \equiv$ `pt-coh` H, π is the calc of `fm-pt-q`). This is recorded as a `refl` lemma `realise-map-def` (rule 3), and Theorem 3 follows.

2.5. **What the canonical map does.** (a) $\mathsf{realise\text{-}map}(a,b,H)\,;p' = p\,;K(a,2)$: `FibMap.fm-over`. Hence on π_2 (through p, p') it induces a. (b) On the fibre over the base point it is Φ's F_{v_0}, and $\mathrm{res}(\Phi(K(b,4),H)) = \mathsf{loop}\,K(b,4)$ (Lemma 11 ($\Rightarrow$) with $z = hs_{v_0}$): the restriction to the fibres is the looping of $K(b,4)$, which is $\mathsf{conj}_3 K(b,3)$ (`loopK`); this is how the coefficient b is read off on π_3. (c) Conversely, the inverse sends G to $a = \mathsf{inv2}$ of the extension of $G\,;p'$ along p (Lemma 4(i)) and b to the coefficient of the delooping of the restriction (Lemma 12).

3. Identities, composition, homotopy categories

Definition 15 (`TripleHomotopy.Maps3.mcomp`, `Maps1.mid`)**.** For $m = (a,b,H) : t \to t'$, $m' = (a',b',H') : t' \to t''$: $m\,;m' = (a\,;a',\ b\,;b',\ H'')$ with H'' the composite

$$K(a\,;a',2)\,;k'' = K(a,2)\,;K(a',2)\,;k'' = K(a,2)\,;k'\,;K(b',4) = k\,;K(b,4)\,;K(b',4) = k\,;K(b\,;b',4),$$

i.e. $\mathsf{ap}_{-;k''}$ of `kc-lev` at 2 (`KFun.Comp.k-comp` through the locks `Lev.K-def`), then the pasting `Paste.paste` (`ptcomp-assoc`, $\mathsf{ap}_{K(a,2);-}H'$, $\texttt{ptcomp-assoc}^{-1}$, $\mathsf{ap}_{-;K(b',4)}H$, `ptcomp-assoc`), then $\mathsf{ap}_{k;-}$ of $\texttt{kc-lev}^{-1}$ at 4 (`Bridge.bridge`). $\mathrm{id}_t = (1_A, 1_B, H_1)$, H_1 the bridge along $K(1,2) = \mathrm{id}$ and $K(1,4) = \mathrm{id}$ (`KId.k2-id`, `k4-id`) of $\texttt{ptcomp-idl} \cdot \texttt{ptcomp-idr}^{-1}$ (`FmId.idsq`). $K(1,m) = \mathrm{id}$ for $m = 2, 4$ is proved from the surjectivity of $K(-,m)$: if $K(h,m) = \mathrm{id}$ then $K(1,m) = K(1,m)\,;K(h,m) = K(1\,;h,m) = K(h,m) = \mathrm{id}$.

Proposition 16 (`rmap-comp-pt`, `rmap-comp`, `rmap-id-pt`, `rmap-id`)**.** $\mathsf{realise\text{-}map}(\mathrm{id}_t)$ *and the pointed identity, and* $\mathsf{realise\text{-}map}(m\,;m')$ *and* $\mathsf{realise\text{-}map}(m)\,;\mathsf{realise\text{-}map}(m')$*, have pointwise equal underlying functions; hence they are merely equal as pointed maps (`pi-mere`,* $\Omega\mathrm{Fib}k''$ *being* $\mathsf{isConn}\,2$*), i.e. equal in* $[\mathrm{Fib}k, \mathrm{Fib}k'']_*$.

Proof. Write $hp(r)_x = \texttt{happly}(\mathsf{ap}_{(-)_1} r)_x$ for the underlying homotopy of a path r of pointed maps (`hp`). By J: $hp(r \cdot s) = hp(r) \cdot hp(s)$, $hp(r^{-1}) = hp(r)^{-1}$, $hp(\mathsf{ap}_{u;-}s)_x = hp(s)_{u_1 x}$, $hp(\mathsf{ap}_{-;v}s)_x = \mathsf{ap}_{v_1}(hp(s)_x)$; $hp(\texttt{ptcomp-assoc}) = \mathsf{refl}$ (J on the three pointings, as in its definition); a path changing only the pointing, such as `ptcomp-idl`, `ptcomp-idr`, has trivial homotopy (`hp-pair`).

(1) `Bridge.fm-bridge`: for paths $U : u_2 = u$, $W : v_2 = v$ of pointed maps and a square s for (u, v), `fm` of the bridged square for (u_2, v_2) is pointwise `fm` of s (J on U, W, whose left endpoints are free; at refl the bridge is $\mathsf{refl} \cdot s$, `trans-refl-l`). (2) `Paste.fm-paste`: the first components agree by definition; the homotopy of the pasting is $hs'_{u_1 x} \cdot \mathsf{ap}_{v'_1}(hs_x)$ (`hs-paste`, the rules above), and the second components agree by `ap-trans`, `ap-ap` and the groupoid identity `grp4` (`eq2`). (3) `FmId.fm-id`: the homotopy of `idsq` is refl, and $\mathsf{refl} \cdot \mathsf{ap}_{\mathrm{id}} q = q$. Then `rmap-comp-pt` is (1) followed by (2), `rmap-id-pt` is (1) followed by (3). □

A coherent pointed path (the base-point coherence) is *not* constructed; the homotopy category only needs classes (as confirmed by the user, this suffices for David Wärn's argument).

Theorem 17 (`TripleHomotopy.Cat`)**.** *Put* $\hom(t, t') = \|\mathsf{Mor}(t, t')\|_0$ *(`homT`) and* $R = \|\mathsf{realise\text{-}map}\|_0 : \hom(t, t') \to [\mathrm{Fib}k, \mathrm{Fib}k']_*$ *(`R`), a bijection by Theorem 3 and `classes-equiv` (`R-equiv`). Composition and identities on* hom *are transported from `compC`, `idC` along R (`compT`, `idT`; R preserves them: `R-comp`, `R-id`), and they* are *the intrinsic ones on representatives:* $[m]\,;[m'] = [m\,;m']$ *and* $\mathrm{id} = [\mathrm{id}_t]$ *(`compT-rep`, `idT-rep`, by Proposition 16 and injectivity of R). The category laws hold (`compT-assoc`, `compT-idl`, `compT-idr`, from `comp-assoc`, `comp-idl`, `comp-idr`). So* $(\mathsf{realise}, R)$ *is a functor from the triple homotopy category to the homotopy category of pointed simply connected* 3*-types which is an equivalence on object types (`PostnikovClassification.classification`; object types are not sets, and no univalence of either category is claimed) and bijective on hom-sets: an equivalence of homotopy categories, with inverse* $\mathsf{realise}^{-1}$ *on objects and* R^{-1} *composed with transport along* $\mathsf{realise}(\mathsf{realise}^{-1} X) = X$ *on hom-sets (the latter is stated here, not formalised as a separate declaration).*

4. The components of the compatible homotopies

Both sphere arguments compute $\hom(t_0, t')$ directly from $\mathsf{Mor}(t_0, t') = \sum_{a,b}(K(a, 2)\,; k' = \kappa_0\,; K(b, 4))$, κ_0 the source invariant: the coefficient b is determined by a through (H4) (with Statement T for $\bar{k}$), and the components of the identity type are controlled by (H3). The latter needs loops at an arbitrary point of $M = \mathrm{Map}_*(K(\mathbb{Z}, 2), Y')$, $Y' = K(B', 4)$; we reduce them to loops at the base point by translation.

Lemma 18 (translation; `TripleComponents.PtTransl`)**.** *Let X, Z be pointed and $M' = \mathrm{Map}_*(X, \Omega Z)$. For $G : M'$, the map $\mathrm{tr}_G(F) = (x \mapsto F_1 x \cdot (G_1 x)^{-1}$, pointing from $F_0, G_0)$ is an equivalence $M' \to M'$ (inverse: right multiplication by G), and $\mathrm{tr}_G(G)$ is merely equal to the constant map* 0 *when $\Omega(\Omega Z)$ is* isConn 2 *(`pi-mere`; pointwise `trans-sym-r`).*

Lemma 19 (`TripleComponents.Unique.unique`, `unique-pt`)**.** *Let X be pointed, Y' pointed with a pointed equivalence $e : \Omega Z \simeq_* Y'$ ($Z = K(B', 5)$, $e = e_4$), $\Omega^2 Z$* isConn 2*, and assume every pointed map $X \to_* \Omega^2 Z$ is merely constant. Then for $f, g : \mathrm{Map}_*(X, Y')$ and $H, H' : f = g$, H and H' are merely equal.*

Proof. Postcomposition with e^{-1} is an equivalence $\mathrm{Map}_*(X, Y') \simeq M'$, so ap of it is injective; then $\mathsf{ap}\,\mathrm{tr}_{e^{-1} g}$ (Lemma 18) reduces to two paths $r, r' : u = v$ in M' with v merely 0; the goal being a proposition, $v = 0$ by J, and $r^{-1} \cdot r'$ is a loop at 0 in M', i.e. (`CupSquare.PtMapLoops`) a pointed map $X \to_* \Omega^2 Z$, merely constant by hypothesis; so $r^{-1} \cdot r'$ is merely refl. □

For $X = K(\mathbb{Z}, 2)$: $\Omega^2 K(B', 5) \simeq_* K(B', 3)$ ($\mathsf{loop}\, e_4, e_3$) and (H3) (`H3Proof.h3`: $[K(\mathbb{Z}, 2), K(B', 3)]_*$ is a point) give the hypothesis.

5. $\pi_3(S^2)$ through triple morphisms

Let $t_0 = (\mathbb{Z}, \mathbb{Z}, \kappa)$, $P = \mathrm{Fib}\kappa$ (`CupSquare.kappa`), and $t' = (A', B', k')$ arbitrary.

Proposition 20 (`Pi3TripleUniversal.ThetaGen.theta-equiv`, `Theta`). $\theta : \hom(t_0, t') \to \mathrm{Hom}(\mathbb{Z}, A')$, $[m] \mapsto$ *mor-a* m, *is a bijection.*

Proof. $\mathrm{Hom}(\mathbb{Z}, A')$ is a set, so θ is well defined. *Surjective:* for a, (H4) (`H4Proof.h4`: $b \mapsto [\kappa\,;K(b,4)]$ is a bijection $\mathrm{Hom}(\mathbb{Z}, B') \to [K(\mathbb{Z},2), Y']_*$) gives b with $[\kappa\,;K(b,4)] = [K(a,2)\,;k']$, hence merely a path H and an element (a, b, H^{-1}) over a. *Injective:* if `mor-a` $m =$ `mor-a` $m' = a$ then $[\kappa;K(b,4)] = [K(a,2);k'] = [\kappa;K(b',4)]$, so $b = b'$ by (H4); transporting, $H, H' : K(a,2);k' = \kappa\,;K(b,4)$, merely equal by Lemma 19 with (H3); so $[m] = [m']$. □

Proposition 21 (`Pi3TripleUniversal.Final.At.marked`). *Let* $s =$ *MapFib.Final.s* $\in [S^2, P]_*$ *(*$s\,;[p] = [\sigma]$*,* *s-eq* *). For every triple* t', $s^* : [P, \mathrm{Fib}k']_* \to [S^2, \mathrm{Fib}k']_*$ *is a bijection.*

Proof. The composite

$$\hom(t_0, t') \xrightarrow{\|\mathsf{realise\text{-}map}\|_0} [P, \mathrm{Fib}k']_* \xrightarrow{s^*} [S^2, \mathrm{Fib}k']_* \xrightarrow{p'_*} [S^2, V']_*$$

sends $[m]$ to $[\sigma\,;K(a,2)]$: $s\,;[\mathsf{realise\text{-}map}\,m]\,;[p'] = s\,;[p]\,;[K(a,2)]$ (`fm-over`, `comp-assoc`) $= [\sigma]\,;[K(a,2)]$ (`s-eq`). The map $a \mapsto [\sigma\,;K(a,2)]$, $\mathrm{Hom}(\mathbb{Z}, A') \to [S^2, V']_*$, is a bijection: σ^* on classes into the 2-type V' (`MapFib.SourceS.Sig.sigma-cls`) after $\|K(-,2)\|_0 : \mathrm{Hom}(\mathbb{Z}, A') \simeq [K(\mathbb{Z},2), V']_*$ (Lemma 14, `classes-equiv`). So the composite is this bijection after θ; $\|\mathsf{realise\text{-}map}\|_0$ (Theorem 3, `classes-equiv`) and p'_* (the lifting lemma at the source S^2, `MapFib.SourceS.qS`, which uses only `s2-paths`) are bijections, hence so is s^* (2-out-of-3). □

Then for $X : \mathsf{SC3}$, $X = \mathsf{realise}(\mathsf{realise}^{-1}X)$ (`classification`), s^* is bijective at X (transport), which is `Pi3Sphere`'s `MapThm`; `Pi3Sphere.Conditional.pi3-S2-T` (truncation, two-map Yoneda, `HomotopyGroups`) gives $\pi_3(S^2) \cong \mathbb{Z}$: `Pi3TripleUniversal.Final.pi3-S2`. *Not used:* `MapFib.Final.rep-fib`, `MapFib.Lifting` at the source P, `FibreH3.fib-h3`, `Presentation`. *Used and disclosed:* the marked class `MapFib.Final.s` (from `qS` at S^2 and σ, not from `rep-fib`).

6. $\pi_4(S^3)$ through triple morphisms and the looping embedding

$\mathsf{SC4}$ embeds in $\mathsf{SC3}$ by Ω (`DeloopEmbedding.Emb.emb`, `sc4-equiv`); on maps, looping is a bijection $[X, Y]_* \simeq [\Omega X, \Omega Y]_*$ for X $\mathsf{isConn}\,4$, Y $\mathsf{isTrunc}\,6$ (`MarkLoop.LoopCls.equiv`), and statement M at $u : \mathsf{SC4}$ is equivalent to bijectivity of $r^* : [R, \Omega Y]_* \to [\Omega S^3, \Omega Y]_*$ with $r = \mathsf{loop}\,s\,;[\omega_{BR}]$ (`MarkLoop.LMark.r`, Proposition L: `LMark.At.from-loop`). Here $\bar{t} = (\mathbb{Z}, \mathbb{Z}/2, \bar{k})$, $R = \mathrm{Fib}\bar{k} = \mathsf{realise}\bar{t}$, $\bar{p} : R \to K(\mathbb{Z},2)$.

Proposition 22 (`Pi4TripleUniversal.Theta4.TG.theta-equiv`). *For every deloopable* $t' = (A', B', k')$ *(*`DeloopEmbedding.Deloopable`*),* $\theta : \hom(\bar{t}, t') \to \mathrm{Hom}(\mathbb{Z}, A')$, $[m] \mapsto$ *mor-a* m, *is a bijection.*

Proof. Surjective: by (H4), $[K(a,2)\,;k'] = \Phi_{B'}(h)$ for some h; by Statement T (`TwoTorsion.TwoTors.Two.two-tors` with $G = K(a,2)$, the only use of deloopability) $2h(1) = 0$, so $h = \rho\,;g$ (`Necessity.Factor`) and $[K(a,2)\,;k'] = \Psi(g) = [\bar{k}\,;K(g,4)]$ (`k-comp`, `ptcomp-assoc`); so merely (a, g, H). *Injective:* for m, m' over a, $\Psi(b) = [K(a,2)\,;k'] = \Psi(b')$, so $b = b'$ (`Necessity.Psi`, injective); the homotopies are merely equal by Lemma 19 with (H3). □

Proposition 23 (`Pi4TripleUniversal.Final4.marked`). *For every* $u : \mathsf{SC4}$*, M holds at* u*.*

Proof. Let $t' = \mathsf{realise}^{-1}(\Omega u)$, deloopable by u, $\varepsilon : \Omega Y \simeq_* \mathrm{Fib}k'$. *Lifting at the source* ΩS^3*, without* `MarkDirect.ExR`: we use `MapFib.Lifting` at $Q = \Omega S^3$ with hypothesis (U) from `MarkDirect.Direct.UnS` (sphere side) and hypothesis (E) proved anew: for $F : \Omega S^3 \to_* V'$, by r^* bijective into $K(A',2)$ (`MarkDirect.Direct.rA`), $\bar{p}^*$ bijective (`MapFib.FibP[`$\mathbb{Z}, \mathbb{Z}/2, \bar{k}$`].PStar`) and `k2-surj`, $[F] = r\,;[\bar{p}\,;K(a,2)]$ for some a; by Proposition 22 (surjectivity) there is merely (a, b, H), and then

$F\,;k'$ is merely $r\,;\bar{p}\,;\bar{k}\,;K(b,4)$, merely 0 (`FibP.fseq` at $\bar{t}$). Hence $p'_*:[\Omega S^3,\mathrm{Fib}k']_*\to[\Omega S^3,V']_*$ is bijective. Then the composite

$$\hom(\bar{t},t')\xrightarrow{\|\mathsf{realise\text{-}map}\|_0}[R,\mathrm{Fib}k']_*\xrightarrow{r^*}[\Omega S^3,\mathrm{Fib}k']_*\xrightarrow{p'_*}[\Omega S^3,V']_*$$

sends $[m]$ to $r;[\bar{p}];[K(a,2)]$ (`fm-over`), and $a\mapsto r;[\bar{p};K(a,2)]$ is a bijection $\mathrm{Hom}(\mathbb{Z},A')\to[\Omega S^3,V']_*$ (Lemma 14, `PStar`, `rA`). So r^* is bijective at $\mathrm{Fib}k'$ (2-out-of-3), at ΩY (`MarkDirect.Direct.transfer` along ε^{-1}), and M holds at u (Proposition L). □

Then $\pi_4(S^3)\cong\mathbb{Z}/2$ as in `Pi4Direct.Assemble`, with *both* instances s_P ($u=BR$) and s_T ($u=\|S^3\|_4$) from this proposition, `Pi4Sphere.Two4` and `Pi4Sphere.pi4-BR`: `Pi4TripleUniversal.Final4.pi4-S3`.

- *Not used:* `MarkDirect.Direct.rF`, `marked`, `qS`, `ExS`; `MarkDirect.ExR`, `qR`; `DeloopEmbedding.Mark.marked`; `SourceS3.Final.rep-fib`; `Presentation4`; `MapBR.ur`.
- *Used and disclosed:* the marked classes `SourceS3.Final.s` and `MarkLoop.LMark.r` (from `qS3` and σ_3); `MarkDirect.Direct.UnS`; `MarkDirect.Direct.rA` (through `SourceS3.Final.right`: σ_3^* on classes into $K(A,3)$); `MarkDirect.Direct.transfer`.

7. Universes

For fixed groups, all of $\mathsf{Mor}(t,t')$, T, S, $\mathrm{Map}_*(\mathrm{Fib}k,\mathrm{Fib}k')$ and $\hom(t,t')$ live in U_l; the types of objects $\mathsf{Triples}$, $\mathsf{SC3}$, $\mathsf{SC4}$ live in U_{l+1}. No universe change is needed in the new modules. In particular, the lemma `ClassBase.conn-lift`, which turns a connectivity statement made with truncations in U_l into the same statement in U_{l+1} and which the object classification needs, is not used: Theorem 3 is stated for fixed $t,t':\mathsf{Triples}$, and in Sections 5–6 the transport along $\mathsf{realise}(\mathsf{realise}^{-1}X)=X$ is a path in $\mathsf{SC3}:U_{l+1}$ acting on statements in U_l, as in `Pi3Classified`.

8. Declaration correspondence

Definitions 1, 2	`TripleMapsEquiv.Maps2.Mor`, `mor-a/b/H`, `ck`, `kc`, `realise-map`, `realise-map-def` (`refl`)
$T(c)$, Definition 5	`TripleMaps.Tc`; `TripleMaps.Key.S`, `T`, `G`, `hs`, `Phi`
Lemma 7	`TripleMaps.PtR`, `ptpath`, `PtPath.equiv`
Lemma 8	`TripleMaps.rmul-equiv`, `Transl.equiv`, `Transl.prop`; `Key.Tl`, `tl-prop`
Lemma 9	`Key.m2-equiv`, `fac-map`, `fac-equiv`
Lemma 11	`Key.res`, `grp1`, `grp3`, `to-loop-at`, `to-loop`, `zt`, `goal-at`, `from-loop`; `res-phi`
Lemma 12	`Key.unique-fac`
Lemma 6	`TripleMaps.Key.key`
Lemma 4	`TripleMapsEquiv.Dec.piq`, `dec`, `piq-equiv`, `cur-equiv`, `bmap-equiv`, `dec-equiv`
Lemma 14	`TripleMapsEquiv.Coef.unconj-conj`, `inv2-K`, `inv3-K`, `inv4-K`, `k2-equiv`, `k4-equiv`
Theorem 3	`TripleMapsEquiv.Maps2.realise-map-equiv`
Section 3	`TripleHomotopy`: `hp*`, `Bridge`, `Paste`, `FmId`, `kc-lev`, `KId`, `Maps3`, `Maps1`, `Cat`
Lemmas 18, 19	`TripleComponents.PtTransl`, `Unique`, `unique-pt`, `const-pt`
Section 5	`Pi3TripleUniversal.H3U`, `KCls`, `ThetaGen` (Proposition 20 for any source triple $(\mathbb{Z}, C, k_s)$ from existence and uniqueness of b), `Theta`, `Final.At.marked`, `Final.pi3-S2`
Section 6	`Pi4TripleUniversal.Theta4`, `Final4.At` (`chain`, `null1`, `ExS4`, `qS4`, `rr-equiv`, `rFt`), `Final4.AtU`, `marked`, `Assemble`, `Final4.pi4-S3`

9. Validation

Fresh checks (28 October 2026): `./segcsag/pi3-s2-postnikovall`: 102 modules, 0 failed; a new process of `caballist-binexe:seg` on `All.seg`: 5633 declarations, 0 failed, 3.4 s. The uniform checker (`seg/headlinear`, `segd--opens`) passes all 32 `.opens` specifications (the 26 existing ones and `triplemaps`, `triplemapsequiv`, `triplecomponents`, `triplehomotopy`, `pi3triple`, `pi4triple`), every entry "nothing else opened". The six new specifications were generated from `--opened` and audited against this note by kind (listed in their headers), not entry by entry; no construction of $K(h, m)$, e_m, the deloopings or the cup square is opened by the new declarations.

Dependency test. In a scratch copy (not the working tree) the old specialised results were deleted: `MapFib.Final.rep-fib`, `rep-pteq`, `rep-cup`, `SourceP.qP`, `ExP`, `ex-P`; `MarkDirect.ExR`, `ex-R`, `un-R`, `qR`, `Direct.ExS`, `ex-S`, `qS`, `rF`, `AtU`, `marked`; `DeloopEmbedding.Mark.At`, `Mark.marked`; `SourceS3.Final.rep-fib`, `rep-pteq`; `FibreH3.fib-h3`, `FibAt`; `Presentation.Final`, `Presentation4.Final4`, `Pi4Sphere.Assemble4`, `Pi4Sphere.Pi4S3`, and the files `Pi3Classified`, `Pi4Direct`. The copy checks with all new modules (5300 declarations, 0 failed). The old endpoints are unchanged in the working tree and still check.

10. What is not claimed

A coherent pointed path for Proposition 16 (only equality in $[\mathrm{Fib}k, \mathrm{Fib}k'']_*$); any bicategorical coherence; univalence of the categories; an explicit inverse functor as a declaration (only the object equivalence and the hom-set bijections with functoriality); a separate formal treatment of the

example $t = (A, 0, 0)$, $t' = (0, C, 0)$; a formal statement that realise-map induces b on π_3 through `Triple.pi3-iso` (what is formalised is `fm-over`, which gives a on the base, and `Key.res-phi`: the restriction to the fibre over the base point is loop $K(b, 4)$).

11. History

A first version was written before any code (`seg/WRITING.md`, rule 1) and refereed independently against the library. It was then revised so that it is the proof checked. The changes, made either after review or during formalisation, are: the π_4 lifting at ΩS^3 (existence now from the surjectivity of θ, instead of `MarkDirect.Direct.qS`, which depended on `MarkDirect.ExR`); the direct computation of the hom-sets from Mor, with the translation and components lemmas (Section 4) replacing an argument through `FibreH3.fib-h3` and `MapBR.ur`; right multiplication (`rmul-equiv`) and `ClassBase.ap-equiv` in Lemmas 8 and 4; the groupoid steps of Lemma 11 ($\Leftarrow$), now `goal-at`; $K(1, m) = \mathrm{id}$ from the surjectivity of $K(-, m)$ instead of an induction; composition on hom transported along R, with its agreement with the intrinsic pasting proved (`compT-rep`).